1- What is an exoplanet

- A planet that orbits a star **other than the Sun**.
- Detected indirectly (most common) via effects on the host star (brightness, radial motion) or directly (imaging).
- Observational signatures depend on planet size, orbit, and host-star properties.

2- Features that help detect exoplanets (prioritized)

(Combines your Kepler feature importance notes and time-series features from tsfresh.)

Highest impact

- **Signal-to-noise (koi_model_snr)** نسبة الإشارة الى الضوضاء— strong single strongest predictor.
- **Transit depth (koi_depth) & duration (koi_duration)** عمق العبور ومدة العبور— shape and depth distinguish planet transits from eclipsing binaries / artifacts.
- **Orbital period (koi_period)** فترة المدار— consistent periodicity is a hallmark of planets.
- **Planet radius estimate (koi_prad)** تقدير نصف قطر الكوكب— extremely large radii often point to false positives (stellar companions).
- **Host star properties (koi_steff, koi_srad, koi_slogg)** خصائص النجم المضيف— used to scale transit depth into physical radius and remove impossible solutions.

Medium / supportive

- **Stellar brightness (koi_kepmag)** سطوع النجم— brighter targets → less photon noise.
- **Impact parameter (koi_impact)** معامل التأثير او الارتطام— refines geometry.
- **Insolation / equilibrium temp (koi_insol, koi_teq)** الإشعاع المستلم / درجة الحرارة التوازنية— contextually useful (habitability) but less for raw detection.
- **Time-series features (tsfresh)** ميزات السلاسل الزمنية— absolute energy, number of values above mean, DFT coefficients, auto-correlations, etc., which capture variability patterns not encoded by domain features.

Vigilance checks (vetting features)

- Odd/even depth comparison, centroid shifts, presence/absence of a secondary eclipse, transit shape (V vs U), and multi-sector consistency.

3- Short history (very condensed)

- **1992** — first confirmed planets discovered around a pulsar (Wolszczan & Frail).
- **1995** — first planet around a Sun-like star (51 Pegasi b: Mayor & Queloz).
- **2000s** — radial velocity (RV) and early transit detections expand the census.
- **2009** — Kepler mission launches → exponential increase in small planet detections.
- **~2014** — Kepler → K2 (campaigns) after reaction wheel issues; new systematics.
- **2018** — TESS launched → all-sky bright-star survey, complementary to Kepler/K2.

4- Mechanisms of identifying exoplanets

Main methods

- **Transit photometry** — measure periodic dips in stellar brightness; your light-curve + BLS/ML pipeline fits here.
    - *BLS (box-fitting least squares)*: classic for periodic box-shaped transits; **limitations**: sensitive to cadence & SNR; vulnerable to cosmic-ray spikes, stellar variability, and random noise → false positives.
    - *ML approaches*: extract engineered features (tsfresh + GBT) or use deep learning (CNNs, RNNs) that automatically learn features from raw curves.
- **Radial velocity (RV)** — measure Doppler shift of star; complementary (gives mass).
- **Direct imaging** — for wide, bright planets (hard).
- **Microlensing, timing variations, astrometry** — niche but powerful in special cases.

Where Our approach fits

- Our pipeline is a **classical ML (feature engineering) + tree-based classifier** approach: robust, interpretable (feature importance), and data-efficient compared to deep learning. Deep learning can automatically learn subtle features but often needs larger labeled datasets and is less interpretable.

5- Influencers / major contributors

- **Alex Wolszczan** — first confirmed exoplanets (pulsar planets, 1992).
- **Michel Mayor & Didier Queloz** — discovered first planet around Sun-like star (51 Pegasi b, 1995); foundational figures.
- **William Borucki** — Kepler mission principal investigator (pioneered large-scale transit surveys).
- **Natalie Batalha** — Kepler science team lead and key in cataloguing Kepler candidates.
- **Sara Seager** — theory of exoplanet atmospheres and observational strategies.
- **Timothy Morton** — statistical validation / false-positive probability tools (e.g., vespa) — important for vetting candidates.

6- Quick comparison: Kepler vs K2 vs TESS

Kepler

- **Strategy**: deep stare on a single field → long baselines.
- **Cadence**: long-cadence ~29.4 min (from research), many (~70k) points per long mission light curve.
- **Strengths**: extremely precise photometry → great for small, long-period planets and robust training data.
- **Systematics**: kepler pipeline corrections; well-characterized catalogs (Autovetter etc).

K2

- **Strategy**: sequential campaigns across ecliptic fields (shorter campaigns than original Kepler).
- **Differences**: more pointing-drift systematics (roll motion), shorter baseline per campaign → more noise/irregularities.
- **Use in your notes**: used as a baseline with injected transits to test detection under realistic systematics.

TESS

- **Strategy**: all-sky survey targeting **bright** nearby stars in short sectors (~27 days) — lots of bright targets.
- **Cadence**: sectors + shorter baselines per field (but multiple sectors near ecliptic poles). Offers short-cadence modes for selected targets.
- **Strengths**: finds planets around bright stars (easier follow-up), good for short-period planets; lower per-target baseline limits long-period sensitivity.
- **Implication for ML**: TESS data → different noise profile and cadence; models trained on Kepler may need domain adaptation (retrain/fine-tune or use injected transits).

7- Summary of the pipeline & experimental choices – from researches

- **Data flow**: raw light curves → clean (remove systematics, cosmic rays, variability) → resample uniformly to **1-hour** → feature extraction (tsfresh → ~790 features) → remove constant features, impute, robust scale → train **gradient-boosted tree (GBT)** classifier.
- **Training / datasets**: simulated K2 baseline (with injected transits) → trained on Kepler → finally TESS photometry. Kepler training labels from Autovetter catalog (PC vs non-PC). Example Kepler counts you used: **3600 PC** and **12,137 non-PC** (80/10/10 split; used same test set as Shallue & Vanderburg for comparison; 10-fold CV across train+val).
- **Performance**: threshold p=0.46 → **recall 0.96, precision 0.82** (comparable to Shallue & Vanderburg 2018).
- **Visualization**: t-SNE used to embed ~700 processed features for cluster visualization and diagnostics.

8- Strengths, weaknesses & practical ML tips (actionable)

Strengths of feature-based GBT approach

- Interpretable (feature importance, SHAP).
- Data-efficient vs deep models.
- Easier to debug and integrate domain knowledge.

Weaknesses / failure modes

- May miss subtle features deep nets can learn (shapes/noise patterns).
- Sensitive to feature redundancy and noisy features → prune constants and correlated features.
- BLS + classical methods vulnerable to cosmic rays, sampling gaps, stellar variability → robust preprocessing needed.

Practical tips

- **Injection-recovery tests**: inject synthetic transits into real light curves to measure sensitivity and completeness.
- **Handle class imbalance**: use class weights, focal loss (if DL), or oversample minority class.
- **Threshold tuning with PR curve**: prefer precision-recall for imbalanced data; select threshold by the operational tradeoff you want (do you prefer recall or precision?).
- **Add vetting features**: centroid shift, odd/even depths, secondary eclipse tests, multi-sector consistency.
- **Model explainability**: use SHAP or feature permutation to justify candidates for judges.
- **Domain adaptation**: fine-tune or retrain for TESS/K2 noise profiles; consider mixing datasets during training.
- **Use ensemble**: combine GBT (fast, interpretable) with a small CNN on folded light curves to capture shape nuances.